

## Problem

For the circuit in Fig. 6.44, the voltage divider formula is:

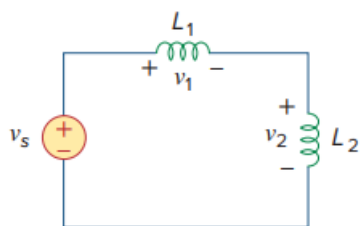
a.  $v_1 = \frac{L_1 + L_2}{L_1} v_s$

b.  $v_1 = \frac{L_1 + L_2}{L_2} v_s$

c.  $v_1 = \frac{L_2}{L_1 + L_2} v_s$

d.  $v_1 = \frac{L_1}{L_1 + L_2} v_s$

**Fig. 6.44:**



## Step-by-step solution

## Step 1 of 4

Refer to figure 6.10 for the circuit in the textbook.

Assume that current  $i$  is flowing in the circuit.

Apply Kirchhoff's voltage law.

$$-v_s + v_1 + v_2 = 0$$

$$v_s = v_1 + v_2 \dots\dots (1)$$

## Step 2 of 4

The voltage across the inductor  $L_1$  is,

$$v_1 = L_1 \frac{di}{dt}$$

The voltage across the inductor  $L_2$  is,

$$v_2 = L_2 \frac{di}{dt}$$

## Step 3 of 4

Recall equation (1).

$$v_s = v_1 + v_2$$

Substitute  $L_1 \frac{di}{dt}$  for  $v_1$ , and  $L_2 \frac{di}{dt}$  for  $v_2$ .

$$\begin{aligned} v_s &= L_1 \frac{di}{dt} + L_2 \frac{di}{dt} \\ &= (L_1 + L_2) \frac{di}{dt} \end{aligned}$$

Therefore,

$$\frac{di}{dt} = \frac{v_s}{L_1 + L_2}$$

#### Step 4 of 4

The voltage across the inductor  $L_1$  is,

$$v_1 = L_1 \frac{di}{dt}$$

Substitute  $\left( \frac{v_s}{L_1 + L_2} \right)$  for  $\frac{di}{dt}$ .

$$\begin{aligned} v_1 &= L_1 \left( \frac{v_s}{L_1 + L_2} \right) \\ &= \left( \frac{L_1}{L_1 + L_2} \right) v_s \end{aligned}$$

Therefore, the voltage divider formula is,

$$v_1 = \left( \frac{L_1}{L_1 + L_2} \right) v_s.$$

Therefore, option **(d)** is correct.